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How can marine ecosystem services support the Blue Growth agenda?

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ABSTRACT

The EU Blue Growth Agenda targets maritime economic activities that have the sea and the coasts as drivers. These activities are supported by marine Ecosystem Services (ES) in combination, or not, with abiotic outputs from the marine natural capital. This paper analyses Blue Growth activities with regards to the demand and supply of marine ES and Good Environmental Status (GES). The results show that marine provisioning ES support aquaculture and blue biotechnology, while blue energy is supported by marine provisioning ES and by abiotic provisioning, and abiotic provisioning supports extraction of marine mineral resources. Maritime, coastal and cruise tourism is supported by cultural marine ES and cultural settings dependent on marine abiotic structures. All these multi-sectoral economic activities depend on healthy marine and coastal ecosystems that are provided by regulating and maintenance ES combined with the abiotic regulation and maintenance by natural marine physical structures and processes. In order to balance concurrent sectoral interests and achieve sustainable use of marine resources there is the need to consider indicators for demand for ES, which are social and economically driven, and for the supply, which are dependent on ecosystems capacity to provide the required marine ES. Some of the actions foreseeing GES are already anticipated in legislation that underpin Blue Growth, whilst others could benefit from additional regulation, particularly in what concern the exploration and exploitation of marine mineral and biological resources. Blue Growth options require navigating trade-offs between economic, social and environmental aspects.

1. Introduction

The Blue Growth agenda, published in 2012 by the European Commission (EC) [1], sets the opportunities for marine, maritime and coastal sustainable growth, transferring the concept of a 'Green Economy' into the 'Blue World' [2], i.e., the 'Blue Economy'. This new economy of the sea aims to optimize the benefits received from the sustainable development of marine environments [3]. The Blue Growth agenda targets the following marine economic sectors: blue energy, aquaculture, maritime, coastal and cruise tourism, marine mineral resources, and blue biotechnology. In this context, the sea and coasts are drivers of the economy, and therefore support a great diversity of human activities that provide human well-being not only to local populations but also to a set of direct and indirect beneficiaries, namely through the sea-related value chain and jobs [4]. These economic activities depend on marine Ecosystem Services (ES) (i.e., outputs of ecosystems mediated by living processes) in combination, or not, with abiotic outputs from the marine natural capital (i.e., stocks of natural assets from which humans derive a wide range of benefits) (Fig. 1).

The economic activities under the Blue Growth agenda might impact marine environmental status, specifically in what concerns Good Environmental Status (GES), meaning that the marine environment is clean, healthy and productive, as defined in the context of European Union (EU) policies. Thus, EU marine-related Directives establish a set of environmental targets, and the associated indicators, to be considered when evaluating the status of the EU maritime space. In this context, a key research question emerges: how can marine ES and the abiotic outputs from marine and coastal ecosystems support the Blue Growth agenda? To answer this question, this paper will assess the links between the Blue Growth agenda and marine ES: specifically, the links between ES that are supplied by seas and coasts and those demanded by the Blue Growth agenda.

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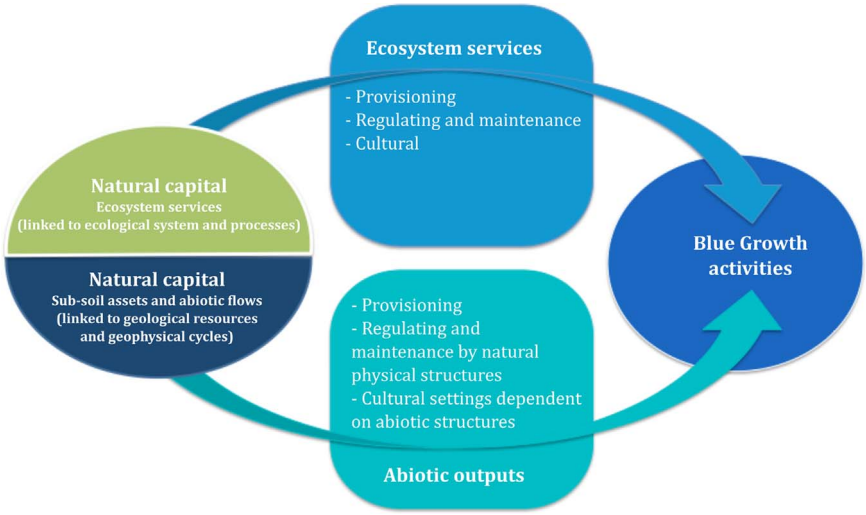

Fig. 1. Schematic representation of how natural capital supports the Blue Growth activities following the Common International Classification of Ecosystem Services (CICES) latest version (V4.3 in [5]). Source: own elaboration from CICES latest version (V4.3 in [6]) and [46].

2. The Blue Growth concept

The concept of Blue Growth perceives the marine space as an environmental and socio-economic system in which there are a set of competing human objectives that need to be addressed in a holistic way (e.g., [5]). The report 'A Blueprint for Ocean and Coastal Sustainability' [2] prepared the discussion for the 2012 Rio + 20 conference and set the way for Blue Growth. The report highlights the oceans broad relevance, referring to oceans as "*a life-support system for human societies*" and the 'Blue-Green' economy as "*key to a sustainable future*", introducing the transition towards a human-ocean centred, but sustainable, relationship. The report identifies a major set of actions to (a) reduce stressors and foster the sustainable use of marine resources and ecosystems, (b) support the 'Blue-Green' economy, and (c) participatory processes to support an effective ocean governance and enable the development of capacities for the sustainable use of the ocean. After the Rio + 20 conference, Blue Growth is seen as an ambitious framework for ocean management that needs to involve stakeholders, to be managed in a goal-oriented approach, and to account for the institutions under which stakeholders make choices [5].

This paper focuses on the 'Blue Economy' in the European maritime space context. However, despite some specificities inherent to EU policies, the principles are applicable to any marine space.

3. Methodology

The present study's approach follows the Common International Classification of Ecosystem Services (CICES) (V4.3 in [6]), but without separating the typologies of marine ecosystems (i.e., it did not separate transitional waters from marine inlets, coastal waters, shelf waters and deep-sea) as proposed by the classification by the EC Working Group on Mapping and Assessment of Ecosystem Services (WG MAES) [7], as this would be too complex for the purpose of this study. Thus, marine ES and abiotic outputs were considered for any marine ecosystem without differentiating marine environments. The present analysis acknowledges that some marine ES are not produced by all the typologies of marine ecosystems considered (e.g., biomass production of seaweed from in situ aquaculture cannot occur in deep-sea environments due to light limitation). However, indicators and associated metrics for biomass production are the same for the relevant marine ES, regardless of typology.

The present analysis focuses on: i) the Blue Growth activities that are linked to the demand (benefits) and/or the supply (production) sides of marine ES; ii) the indicators for trade-off analysis between concurring sectoral-activities and environmental aspects associated with future developments; and iii) how GES and related EU policies link to the Blue Growth agenda.

In order to establish the possible links between marine ES supply, the marine and coastal main economic activities, and the Blue Growth agenda demand for marine ES, the data and information collected was organized into the three sections of marine ES combined with the three sections of abiotic outputs from marine ecosystems. The complete CICES V4.3 spreadsheet that was used in the Marine Pilot on Mapping and Assessment of Ecosystem Services [7,8], i.e., the MAES Matrix, was also used as a starting point for this approach, and is available at http://cices.eu/. The maritime, marine and coastal economic related activities considered in the analysis were the ones included in the Blue Growth agenda [1].

4. Results and discussion

The concept of Blue Growth seeks to realise *"the potential of our seas and oceans for jobs and growth"* [9] following the principles of conservation, of sustainable development, and of management in an innovative way [1,3,10]. Blue Growth also foresees approaches to restore marine and coastal habitats, biodiversity and ES, whilst innovation might support the development of cost-effective protection measures in support of an effective implementation of maritime, marine and coastal-related policies, agreements and international conventions (e.g. Marine Spatial Planning, Integrated Coastal Zone Management, RAMSAR, HELCOM, Barcelona and Black Sea conventions, the UN Convention on Biological Biodiversity and the UN Convention on the Law of the Sea). The discussion around different elements that are part of the question "how can marine ES support the Blue Growth agenda?" is summarized in Fig. 2. This figure shows the links between the demand for ES that are driven by the Blue Growth agenda's economic activities (blue energy, aquaculture, maritime, coastal and cruise tourism, marine mineral resources, and blue biotechnology) and the supply side for ES. In addition, it should be noted that marine ecosystems could be negatively impacted by these economic activities, if such activities are not properly regulated and managed, halting marine ecosystems' capacities to provide the demanded ES. Therefore, research is needed on the impacts from these multiple human activities taking place in the maritime, marine and coastal space on marine ES [11]. In this sense this work establishes the first step to establish the links between marine ES and the economic-related activities included in the Blue Growth agenda.

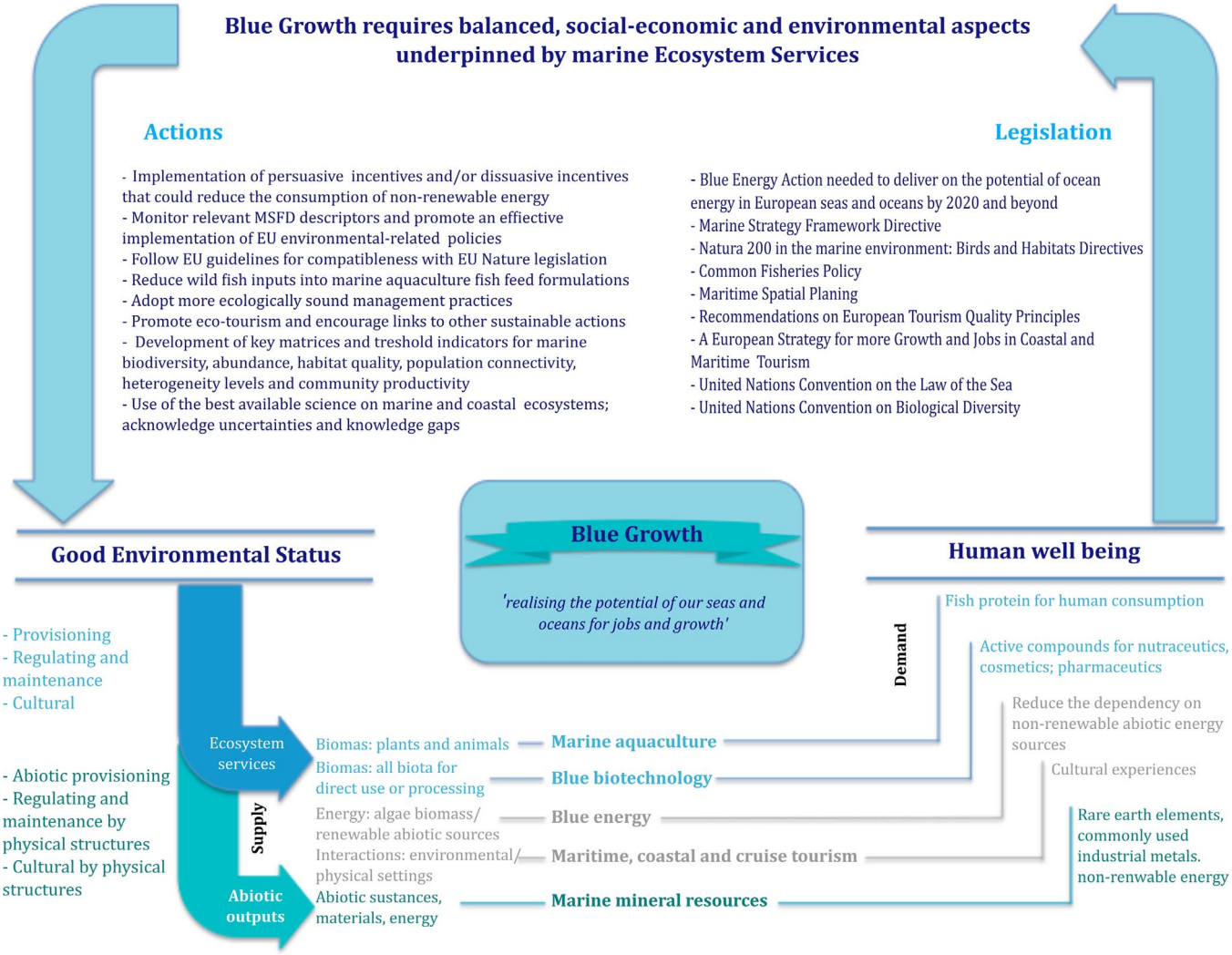

Fig. 2. Schematic representation of how marine ecosystem services (ES) can support the Blue Growth agenda, taking into account the demand for marine ES and the actions regulated by legislation that are needed for the supply of the required services in a sustainable way. Source: own elaboration.

4.1. Provisioning, regulating and maintenance, and cultural ES supporting Blue Growth

Tables 1–3 show links between different types of marine ES and activities foreseen in the Blue Growth agenda. Specifically, Table 1 shows these links for provisioning marine ES and marine abiotic provisioning; Table 2 for regulating and maintenance marine ES and regulation and maintenance by natural marine physical structures and processes; and Table 3 for cultural marine ES and cultural settings dependent on marine abiotic structures.

Abiotic outputs from the sea and the coast's natural capital are as important as ES provided for the marine and coastal economic activities taking place in the maritime space, particularly in what concerns marine provisioning ES and marine abiotic provisioning. Marine provisioning ES support Blue Growth areas of aquaculture (plants and animals from in situ aquaculture for human nutrition) and blue biotechnology (fibbers and other materials from all biota for bio-prospection of active compounds for nutraceutics, pharmaceutics and cosmetics). In the case of blue energy, although it can include the production of biofuel from macroalgae and microalgae supported by marine ES, it is primarily supported by abiotic outputs from marine ecosystems natural capital (wind, wave and solar (WWS) energy resources and non-renewable abiotic energy sources like oil and gas). Marine abiotic provisioning supports the extraction of marine mineral resources (e.g., poly-metallic nodules, cobalt-rich crusts, poly-metallic massive sulphides) to provide for rare earth elements (REEs) and other commonly used industrial metals (e.g., Cr, Ni, Zn, Mo, Pb, W), and for concrete aggregates for construction (e.g., sand and gravel) (Table 1).

Table 2 shows that regulating and maintenance marine ES, and regulation and maintenance by natural marine physical structures and processes, do not directly support the Blue Growth focus areas (i.e., blue energy, aquaculture, maritime, coastal and cruise tourism, marine mineral resources, and blue biotechnology). However, these maritime, marine and coastal-related economic activities depend on healthy marine and coastal ecosystems, i.e. ecosystems in GES, which in turn depends on the structure, functions and processes of marine ecosystems that underpin regulating and maintenance ES combined with the regulation and maintenance by natural marine physical structures and processes.

The EU Marine Strategy Framework Directive (MSFD) [12] follows an ecosystem-based approach and the GES concept integrates the ecosystem structure, functions and processes with physiographic, geographic, biological, geological, chemical, physical and acoustic conditions, and climatic factors, including those resulting from anthropogenic activities. MSFD is the first EU legislative instrument focusing on the protection of marine biodiversity, as it contains the explicit regulatory objective that 'biodiversity is maintained by 2020'. In fact, marine biodiversity is the first of the 11 descriptors defined by the MSFD as mandatory for achieving GES. There are relevant examples of indirect links between the regulating and maintenance of ES combined

Table 1
Summary table with the provisioning marine ecosystem services (ES) and marine ecosystems abiotic provisioning. The first two columns show the divisions and groups, followed by the indicator for supply of ES, the corresponding main ongoing marine and maritime economic activities and the Blue Growth focus areas of activities, and finally the example of an indicator for demand of ES (N.A. stands for not applicable, or service too negligible in magnitude to be worth assessing). Source: own elaboration from CICES latest version (V4.3 in [8]) and [9]. Source: own elaboration from CICES latest version (V4.3 in [8]).

Provisioning marine ES

Division	**Group** (includes several classes)	**Indicator for supply** (metrics)	Marine and maritime economic activities **Blue Growth agenda**	**Indicator for demand** (metrics)
Nutritional	**Biomass** Wild plants and fauna; plants and animals from in situ aquaculture	**Yield** (ton ha^{-1})	Fisheries Marine aquatic products **Marine aquaculture**	**Fish protein for human consumption** (nutritional value)
	Water Surface or groundwater for drinking purposes	**N.A**	**N.A**	**N.A**
Materials	**Biomass** Fibbers and other materials from all biota for direct use or processing; genetic materials (DNA) from all biota	**Yield** (ton ha^{-1})	Marine aquatic products Blue biotechnology **Blue biotechnology**	**Active compounds for nutraceutics; pharmaceutics; cosmetics** (number of licensed patents)
	Water Surface or groundwater for non-drinking purposes	**Volume** (Mm^{-3})	Land-based **Marine aquaculture**	**Water in good chemical quality status** ([O_2], [NH_4-N], [NO_x-N])
Energy	**Biomass** Macro and microalgae	**Yield** (ton ha^{-1})	**Blue energy** produced through **Marine aquaculture**	**Reduce the dependency on non-renewable abiotic energy sources** (kW $h^{-1)}$

Marine ecosystems abiotic provisioning

Division	**Group**	**Indicator for supply** (metrics)	Marine and maritime economic activities **Blue Growth agenda**	**Indicator for demand** (metrics)
Nutritional abiotic substances	**Mineral** Marine salt	**Yield** (ton ha^{-1})	Marine aquatic products	**Essential minerals required for human body to be fit & healthy** (mineral composition)
	Non-mineral Salt-water	**Volume** (m^3)	Desalinization	**Demand for drinking water** (billion m^3 y^{-1})
Abiotic materials	**Metallic** Poly-metallic nodules; cobalt-rich crusts, poly-metallic massive sulphides	**Area** (km^2)	Marine mineral mining **Marine mineral resources**	**Demand for rare earth elements (REEs) & for commonly used industrial metals (e.g. Cr, Ni, Zn, Mo, Pb, W)** (kg m^{-2})
	Non-metallic Sand/gravel	**Yield** (ton ha^{-1})	Aggregates mining **Marine mineral resources**	**Concrete aggregates for construction & other concrete products** (billion ton y^{-1})
Energy	**Renewable abiotic energy sources** Wind, wave and solar (WWS) energy resources	**Energy** (million watts)	Offshore wind Ocean renewable energies **Blue energy**	**Reduce the dependency on non-renewable abiotic energy sources** (kW h^{-1})
	Non-renewable abiotic energy sources Oil and gas	**Volume or weight** (billion barrels or tonnes)	Offshore oil and gas **Marine mineral resources**	**Energy for all purposes (electric power, transportation, heating/cooling, etc.)** (thermal energy units; Kilojoule)

Table 2
– Summary table with the regulating and maintenance marine ecosystem services (ES) and regulation and maintenance by natural marine physical structures and processes. The first two columns show the divisions and groups, followed by the indicator for supply, the corresponding main ongoing marine and maritime economic activities and the Blue Growth focus areas of activities, and finally the indicator for demand (N.A. stands for not applicable, or service too negligible in magnitude to be worth assessing; *stands for indirect link).
Source: own elaboration from CICES latest version (V4.3 in [8]) and [9]. Source: own elaboration from CICES latest version (V4.3 in [8]).

Regulating and maintenance marine ES

Division	**Group** (includes several classes)	**Indicator for supply** (metrics)	Marine and maritime economic activities **Blue Growth agenda**	**Indicator for demand** (metrics)
Mediation of waste, toxics and other nuisances	**Mediation by biota** **Mediation by ecosystems** Combination of biotic and abiotic factors	**Species composition and area covered with wetlands (following Ramsar convention definition)** (stand density ha^{-1})	Coastal protection Monitoring and surveillance Fisheries Marine aquatic products **Marine aquaculture*** **Blue biotechnology*** **Blue energy***	**Water in good chemical quality status** ([O_2], [NH_4-N], [NO_x-N], [priority substances] [potential toxic elements], [emergent contaminants])
Mediation of flows	**Mass flows** **Liquid flows** **Gaseous/air flows**			
Maintenance of physical, chemical, biological conditions	**Lifecycle maintenance, habitat and gene pool protection** **Pest control**	**Habitat health status** (Habitat fragmentation index; presence of alien species)		**Diversity and abundance of species with potential for biotechnology purposes; absence of pathogens** (H', ind. m^{-2})
	Soil formation and composition **Water conditions** **Atmospheric composition and climate regulation**	**Biogeochemical cycles (C, N, P); Metals and POP deposition** N removal (%); C stock (ton C); C sequestration (ton C y^{-1}) [Metals], [POP]		**Water in good physicochemical and microbiological quality status** ([O_2], pH [CO2], temperature, [DIN])

Regulation and maintenance by natural marine physical structures and processes

Division	**Group**	**Indicator for supply** (metrics)	Marine and maritime economic activities **Blue Growth agenda**	**Indicator for demand** (metrics)
Mediation of waste, toxics and other nuisances	**By natural chemical and physical processes** Atmospheric dispersion and dilution; adsorption and sequestration of waters in sediments; screening by natural physical structures	**Adsorption desorption processes; OM sedimentation; Metals and POP deposition** [Metals], [POP] % OM, sedimentation rates	Coastal protection Monitoring and surveillance **Marine mineral resources*** **Marine aquaculture*** **Maritime, coastal and cruise tourism*** **Blue energy***	**Water in good physicochemical and quality status** ([O_2], pH [CO_2], temperature, [priority substances], [potential toxic elements], [emergent contaminants] [POM], [DOM])
Mediation of flows by natural abiotic structures	**By soild (mass), liquid and gaseous (air) flows** Protection by sand and mud flats; topographic control by dunes and cliffs of wind erosion			
Maintenance of physical, chemical, abiotic conditions	**By natural chemical and physical processes** Sea breezes			

Table 3
Summary table with the cultural marine ecosystem services (ES) and cultural settings dependent on marine abiotic structures. The first two columns show the divisions and groups, followed by the indicator for supply, the corresponding main ongoing marine and maritime economic activities and the Blue Growth focus areas of activities, and finally the right the indicator for demand. (N.A. stands for not applicable, or service too negligible in magnitude to be worth assessing; *stands for indirect link).
Source: own elaboration from CICES latest version (V4.3 in [8]) and [9]. Source: own elaboration from CICES latest version (V4.3 in [8]).

Cultural marine ES

Division	**Group** (includes several classes)	**Indicator for supply** (metrics)	Marine and maritime economic activities **Blue Growth agenda**	**Indicator for demand** (metrics)
Physical and intellectual interactions with biota, ecosystems, and seascapes [environmental settings]	**Physical and experiential interactions** Experiential use of biota and seascapes; physical use of seascapes in different environmental settings	**Recreational trips** (no. y^{-1}) **In water recreational activities occurrence** (no.); **Extent of marine protected areas** (ha); **presence of iconic/endangered species** (no.);	Coastal tourism Yachting and marinas Cruise tourism **Maritime, coastal and cruise tourism**	**Number of passengers** (no.) **and the generated income per passenger** (€) **Willingness to pay to keep the supply side** (€)
	Intellectual and representational interactions Scientific; education, heritage; aesthetic; entertainment	**Generated knowledge available in scientific publications** (no.)	Marine aquatic products **Marine aquaculture* Blue biotechnology***	**Applicability of the new knowledge in Marine aquaculture facilities and in the bio-prospection of new active compounds** (no. of patents)
		Seascape beauty estimation (questionnaires; seascape metrics)	**Maritime, coastal and cruise tourism**	**Travel cost estimation** (€); **turnover from tourism** (€/ha*a); **available paintings/ illustrations, songs, products portraying the seascape/ecosystem** (no. ecosystem type)
Spiritual, symbolic and other interactions with biota, ecosystems, and seascapes [environmental settings]	**Spiritual and/or emblematic** Symbolic; sacred and/or religious	**Presence of endangered, protected, iconic and/ or rare species or habitats** (no./ha	**Maritime, coastal and cruise tourism**	**Personal preferences elicited through interviews/questionnaires**
	Other cultural outputs Existence; bequest	**Extent of MPAs** (ha); **Presence of endangered, protected, iconic and/or rare species or habitats** (no./ha)	**Maritime, coastal and cruise tourism**	**Willingness to pay to maintain/preserve the supply side** (€) **personal preferences elicited through interviews/questionnaires**

Cultural settings dependent on marine abiotic structures

Division	**Group**	**Indicator for supply** (metrics)	Marine and maritime economic activities **Blue Growth agenda**	**Indicator for demand** (metrics)
Physical and intellectual interactions with land-/seascapes [physical settings]	By physical and experiential interactions or intellectual and representational interactions	**Extent and variability of coastal seascapes (e.g. fjords, islands, submarine canyons)** (no./ha)	Deep-sea shipping Short-sea shipping Passenger ferry services **Maritime, coastal and cruise tourism**	**Photography's taken and shared** (no.)
Spiritual, symbolic and other interactions with land-/seascapes [physical settings]	By spiritual or symbolic interactions	**Extent and variability of sacred coastal seascapes** (no./ha)	**Maritime, coastal and cruise tourism**	**Personal preferences elicited through interviews/questionnaires; travel cost estimation** (€)

Table 4
The aims of the main marine ecosystems related legislation that regulate the human activities in the marine and maritime environments supporting the optimization of the benefits received from the sustainable development as foreseen in the Blue Growth agenda focus areas, and the main actions to ensure sustainability and Good Environmental Status (GES). Source: own elaboration.

Marine ecosystems-related legislation	Blue Growth	Sustainability and GES
Action needed to deliver on the potential of ocean energy in European seas and oceans by 2020 and beyond **Aim:** mitigate EU dependence on fossil fuels for electricity generation and enhance energy security [24]. **Marine Strategy Framework Directive (MSFD)** **Aim:** hydro-morphological, physical and chemical properties of the ecosystems, including those properties which result from human activities in the area concerned, allow those ecosystems to function fully and to maintain their resilience to human-induced environmental change. Anthropogenic inputs of substances and energy, including noise, into the marine environment do not cause pollution effects [12]. **Maritime Spatial Planning Directive (MSP)** **Aim:** sustainable development of energy sectors at sea [18]. **Natura 2000 in the marine environment: Birds** [28] **and Habitats Directives** [29]. **Aim:** the Habitats Directive does not, a priori, exclude wind farm developments in or adjacent to Natura 2000 sites. These need to be judged on a case-by-case basis [30].	**Blue energy**	**Action:** persuade incentives could ensure that wind, wave and solar energy resources systems develop as quickly and broadly as is socially desirable (e.g. feed-in tariffs (FITs), output subsidies, investment subsidies, output quotas) [25]. **Action:** dissuade incentives could reduce the consumption of non-renewable energy (e.g. by eliminating subsidies for fossil-fuel energy, or taxing fossil-fuel production and use to reflect the expected cost of climate change due to CO_2 emissions [25]. **Action**: mitigation measures as wind farms, platforms and energy installations produce underwater noise that might impact negatively marine ecosystems [26,27]. **Action:** monitor MSFD descriptor 11 on introduction of energy (including underwater noise that is closely linked with pressures due to the offshore wind farms [12]. **Action**: a guidance on how best to ensure that wind energy developments are compatible with the provisions of the Habitats and Birds Directives, namely in what concerns potential impacts of offshore wind farms on marine animals [30].
Common Fisheries Policy (CFP) **Aim:** to ensure the European fishing industry is sustainable and does not threaten the fish population size and productivity over the long term [31]. **Marine Strategy Framework Directive (MSFD)** **Aim:** to achieve GES of the EU's marine waters by 2020 and to protect the resource base upon which marine-related economic and social activities depend [12]. **Maritime Spatial Planning Directive (MSP)** **Aim:** promoting the sustainable growth of maritime economies, the sustainable development of marine areas and the sustainable use of marine resources [18]. **Natura 2000 in the marine environment: Birds** [28] **and Habitats Directives** [29]. **Aim:** Marine Protected Areas (MPAs) as foundation of Ecosystem-Based Management (EBM).	**Marine aquaculture**	**Action:** if the growing aquaculture industry is to sustain its contribution to world protein supplies for human consumption, it must reduce wild fish inputs into fish feed formulations [34]. **Action:** monitor MSFD descriptor 3 on population of commercial fish species is healthy that might be closely linked with pressures due to wild fish inputs into fish feed formulations [12]. **Action:** adopt more ecologically sound management practices [34]. **Action:** contribute to the sustainable development of aquaculture areas, considering economic, social and environmental sustainability, applying an ecosystem-based approach. Maritime spatial plans should promote the coexistence of relevant actual and/future activities and uses of the target area (e.g. [44]). **Action:** a guidance that facilitates the knowledge and implementation of EU legislation underpinning Natura 2000 in relation to aquaculture activities, by promoting best practices which illustrate how nature protection provisions can be compatible with sustainable aquaculture development [37].
COM(2014) 85 final - Recommendation on European Tourism Quality Principles **Aim:** to increase consumer confidence and the reliability of tourism services and products in Europe [38]. **COM(2014) 86 final - A European Strategy for more Growth and Jobs in Coastal and Maritime Tourism** **Aim:** to exploit the tourism sector growth potential in a way that sustainably produces economic benefits [40]. **Maritime Spatial Planning Directive (MSP)**. **Aim:** promotion of sustainable tourism [18] **Natura 2000 in the marine environment: Birds** [28] **and Habitats Directives** [29]. **Aim:** Marine Protected Areas (MPAs), a key mechanism to safeguard biodiversity and increase the resilience of ecosystems to unwanted change [45].	**Maritime, coastal and cruise tourism**	**Action**: effective implementation of EU environmental related policies (e.g. WFD, MSFD, MSP, ICZM) can be seen as a prerequisite for tourism to develop and succeed [38]. **Action:** monitor MSFD descriptor 2 on non-indigenous species do not adversely alter the ecosystem that might be closely linked with baluster waters; and descriptor 10 on properties and quantities of marine litter in the coastal and marine environment [12]. **Action:** promote eco-tourism and encourage links to other sustainability actions; encourage the diversification and integration of coastal and inland attractors, including through transnational thematic itineraries like cultural, religious or ancient trade routes [40]. **Action:** address environmental pressures [24] **Action**: Monitor the type and operational behaviour of cruise vessels, as impacts of both collisions with cetacean species and acoustic pollution are closely linked with type, age, size and speed of a vessel [39]. **Action:** successful management of MPAs, Natura 2000 sites, UNESCO Biosphere reserves provides significant recreational opportunities and contributes to sustainable growth, employment and their conservation for future users.
The United Nations Convention on the Law of the Sea (UNCLOS) **Aim:** to govern and coordinate deep-sea floor issues, in particular deep-sea minerals, through the International Seabed Authority (ISA), an autonomous international organization (http://www.isa.org.jm/authority). **Marine Strategy Framework Directive (MSFD)** **Aim:** the structure, functions and processes of the constituent marine ecosystems, together with the associated physiographic, geographic, geological and climatic factors, allow those ecosystems to function fully and to maintain their resilience to human-induced environmental changes [12]. **Maritime Spatial Planning Directive (MSP)** **Aim:** sustainable extraction of raw materials [18].	**Marine mineral resources**	**Action:** development of key metrics that may serve as threshold indicators are required, such as biodiversity, abundance, habitat quality, population connectivity, heterogeneity levels, and community productivity [43]. **Action:** monitor MSFD descriptor 6 on sea floor integrity ensuring the functioning of the ecosystem that is closely linked with pressures due to the exploitation of marine mineral resources [12]. **Action:** European Parliament funded research on STOA for deep-seabed exploitation: Tackling economic, environmental and societal challenges [41]. **Action:** use of the best available science on marine and coastal ecosystems; acknowledge uncertainties and knowledge gaps. **Action:** policy options are suggested to balance trade-offs between economic, social and environmental aspects associated with future developments [41].
Convention on Biological Diversity (CBD)	**Blue bio-technology**	**Action:** development of key metrics that may serve as threshold

(continued on next page)

Table 4 *(continued)*

Marine ecosystems-related legislation	Blue Growth	Sustainability and GES
Aim: ultimately conservation and sustainable use of biological diversity will strengthen friendly relations among states and contribute to peace for humankind [42]. **Marine Strategy Framework Directive (MSFD)** **Aim:** marine species and habitats are protected, human-induced decline of biodiversity is prevented and diverse biological components function in balance [12]. **Natura 2000 in the marine environment: Birds [28] and Habitats Directives [29].** **Aim:** evaluate the actual performance of specific measures within individual MPAs in terms of their effect upon marine biodiversity (including species biomass, size and species richness) [45].		indicators are required, such as biodiversity, abundance, habitat quality, population connectivity, heterogeneity levels, and community productivity [43]. **Action:** monitor MSFD descriptor 1 on biodiversity that is closely linked with pressures due to harvesting of wild biological resources [12]. **Action:** overcome the lack international legislation to regulate marine biological resources beyond national jurisdiction, by following intervening pieces of legislation such as the Convention on Biological Diversity or UNCLOS may consider implementing a new agreement to include biological and genetic resources [41]. **Action:** use of the best available science on marine and coastal ecosystems; acknowledge uncertainties and knowledge gaps. **Action:** policy options are suggested to balance trade-offs between economic, social and environmental aspects associated with future developments [41].

with the regulation and maintenance by natural marine physical structures and processes to the Blue Growth focus areas. As can be seen in Table 2, the demand for marine water in good chemical quality status: for marine aquaculture, to produce biofuel from macro and microalgae, and for the good habitat health needed to ensure the diversity and abundance of native species with biotechnology potential as well as the absence of pathogens or alien invasive species: are indirectly linked to regulating and maintenance ES. Most of the abovementioned economic activities require specific physical, chemical, and abiotic conditions that depend on the natural abiotic structures, or require specific processes that are mediated by natural marine physical structures within the maritime, marine and coastal space.

Maritime, coastal and cruise tourism is supported by physical, intellectual, spiritual or symbolic interactions related to environmental settings, i.e., cultural marine ES, or related to physical settings, i.e., cultural settings dependent on marine abiotic structures. Marine aquaculture and blue biotechnology activities can also benefit from the supply of the generated knowledge available in scientific publications (cultural marine ES) (Table 3).

4.2. Blue Growth demand and supply indicators

It is well recognised that maritime, marine and coastal-based activities and related economies are often competing for the same space (e.g. shipping and maritime transport, offshore energy, ports development, fisheries and aquaculture and environmental concerns) [13–15] and are often dependent on each other (e.g. [10,16]). At the EU level, the Integrated Maritime Policy [17] fosters a more coherent approach to maritime issues, with increased coordination between different maritime, marine and coastal-related policies, in line with the Europe 2020 goals for an innovative perspective that can unlock new sources of growth and jobs. This cross-sectoral Policy has Maritime Spatial Planning (MSP) [18] as a key tool for improved decision-making in the maritime space [19]. At the global level MSP, as defined by UNESCO "*is a public process of analysing and allocating the spatial and temporal distribution of human activities in marine areas to achieve ecological, economic, and social objectives that usually have been specified through a political process*" [20]. Therefore, MSP can be seen as a practical framework for arbitrating between competing Blue Growth economic activities and the environment. In line with MSP, Blue Growth requires the establishment of a more rational use of marine and coastal space, the interactions among its uses, and the safeguard of the marine environment. In order to balance sectoral interests and achieve sustainable use of marine resources, there is the need to consider demands for social-economic development, e.g., demand for: fish/algae protein for human consumption (marine aquaculture); active compounds for nutraceutics pharmaceutics and cosmetics (blue biotechnology); reduction of the dependency on non-renewable abiotic energy sources (blue energy); REEs and commonly used industrial metals (e.g. Cr, Ni, Zn, Mo, Pb, W) as well as for non-renewable energy (exploitation of Marine mineral resources) (Table 1); and for cultural experiences, i.e., physical, intellectual, spiritual or symbolic interactions with environmental and/or with physical settings (maritime, coastal and cruise tourism) (Table 3), with the need to protect the environment. While the demand side is socio-economically driven by the Blue Growth agenda, the supply side is shaped by ecosystems' capacity to provide the required ES that support human activities.

Tables 1–3 also illustrate some examples of indicators and associated metrics for the demand and supply sides for Blue Growth economic activities and the supporting ES or the abiotic output from the ecosystem. Indicators for demand and supply are of paramount importance to policy options that imply trade-offs between economic, social and environmental aspects associated with future developments in the maritime, marine and coastal space (e.g. [8,11,21]) in order to balance the Blue Growth multi-sectoral interests. Trade-off evaluation and management of these interests will require a concerted effort, and evidence-based ecosystem assessments that incorporate knowledge of ecosystem functioning and services, will help identify appropriate management measures ensuring environmental sustainability and GES [22].

4.3. Blue Growth and Good Environmental Status (GES)

In the context of EU policies that regulate human activities in the marine, maritime and coastal environments, Table 4 establishes links between the aims of the main EU marine ecosystems-related policies, supporting the optimization of the benefits received from sustainable development as foreseen in the Blue Growth agenda, and the main actions to ensure sustainability and GES. Some of the actions foreseeing GES are already anticipated in legislation that underpin the Blue Growth aims, whilst others could benefit from additional actions or regulations. In fact, Jones et al. [23] pointed to the fact that despite the EC's general aim to ensure that policies and initiatives for Blue Growth should be consistent with the achievement of GES, some policies (e.g., MSFD and MSP) have different attributes that might make integration difficult. Marine Protected Areas (MPAs) illustrate this inconsistency. They are considered by the MSFD as indispensable management tools to achieve GES, while regarding the MSP they are just one of the several uses of sea space that can be traded-off against other uses. Such tensions between policies can compromise an effective sustainable use of marine ES, especially given the recent prioritisation by the EC of Blue Growth as a strategy to foster the recovery from the global financial crisis.

Analysing each of the Blue Growth sectoral economic activities [9], marine renewable energy has a dedicated piece of legislation regarding

the actions needed to deliver the potential of ocean energy in European seas and oceans by 2020 and beyond. The proposed actions aim to maximize marine renewable energy sources in order to diminish greenhouse gas emissions, reduce the dependency on non-renewable energy sources and enhance energy security [24].

The development and broad implementation of blue energy sources could be enhanced through a combination of persuasive incentives for wind, wave and solar energy resources (e.g., feed-in tariffs (FITs), output subsidies, investment subsidies, output quotas), with dissuading incentives to the consumption of non-renewable energy (e.g., eliminating subsidies for fossil-fuel energy, or taxing fossil-fuel production and use to reflect the expected cost of climate change due to CO_2 emissions) [25]. Although these blue energy sources are more environmentally friendly, they also pose some risks to the maritime, marine and coastal environment, for example the underwater noise derived from marine wind farms [26,27], or the competition for marine spaces with traditional activities such as small-scale fisheries. So, the use of marine renewable energies should not compromise the GES of marine ecosystems, in particularly the underwater noise as defined by the 11th Descriptor of the MSFD. Therefore, in 2011 the EU published a guidance on how best to ensure that wind energy developments are compatible with EU Nature legislation (Natura 2000 sites: Birds [28] and Habitats [29] Directives), namely in what concerns potential impacts of offshore wind farms on marine animals [30].

Regarding aquaculture, the revised Common Fisheries Policy (CFP) [31] aims to ensure that the European fishing industry is sustainable and does not threaten stock sizes and fisheries yield over the long term. However, the percentage of wild fish catches used to feed aquaculture might not be compatible with the objectives of the CFP [32,33]. In this sense, if the growing aquaculture industry is to sustain its contribution to animal protein supplies for human consumption, the way forward is to reduce wild fish inputs into fish feed formulations [34]. In fact, there are already efforts in this direction, namely amongst the main global aquaculture producers such as China. While aquaculture production in China almost doubled from 2000 to 2014 (increasing from 24.6 million t to 47.5 million t) (China Fishery Statistical Yearbook 2001–2015), the inclusion of fishmeal in feeds for the most important reared species declined considerably [35]. However, it is also important to highlight that China has diversified the sources of fish meal/oil not only by increasing the use of soymeal and oil [33], but also participating and/or buying fishing enterprises where the fishery resource is located[1] as well as by expanding the operation of the Chinese fishing vessels globally [36].

Concerning the interaction between marine aquaculture and conservation, namely MPAs, the EC published in 2012 a guidance document on aquaculture activities in the context of the Natura 2000 network, which promotes best practices and illustrate how nature protection can be compatible with sustainable aquaculture development [37].

For maritime, coastal and cruise tourism, actions foreseeing GES are already anticipated in the legislation that underpin Blue Growth agenda, namely the recommendation on the principles for a quality European tourism [38], which foresee the assessment of environmental pressures and the effective implementation of EU environmental related policies (e.g. WFD, MSFD, MSP, ICZM) as prerequisites for successful tourism; including monitoring the type and operation of cruise vessels, as impacts of collisions with cetaceans and acoustic pollution are closely linked with type, age, size and speed of the vessel [39]. The European strategy to increase growth and jobs in coastal and maritime tourism also foresees actions to promote eco-tourism and encourage links to other sustainable actions. It also encourages diversification and integration of coastal and inland attractors, including through transnational thematic itineraries like cultural, religious or ancient trade routes [40].

Since the beginning of the new millennium the exploration and exploitation of the deep-seas for marine minerals and genetic resources has been a subject of increased interest. The European Parliament published a report on technology options for deep seabed exploitation [41] in which it acknowledges that the future remains uncertain regarding to what extent the seafloor will be exploited on a commercial scale; whilst in terms of blue biotechnology it sees large potential for finding more applications from marine genetic and biological resources. At the global level, the UN Convention on the Law of the Sea (UNCLOS) is the most important international law for governing deep-sea resources. In areas beyond national jurisdiction, the International Seabed Authority (ISA), an autonomous international organization, was set to govern and coordinate deep-sea floor issues, in particular deep-sea minerals. However, managing biological and genetic resources in areas beyond national jurisdiction is not included in the competencies of ISA, as they are restricted to mineral resources (Article 133(a) of the Convention) [41]. As suggested by the Science and Technology Options Assessment (STOA) working group for the European Parliament, this gap could be filled by the Convention on Biological Diversity (1992) [42]. As can be read in the original document, the Convention "*desires to enhance and complement existing international agreements for the conservation of biological diversity and sustainable use of its components and determine to conserve and sustainably use biological diversity for the benefit of present and future generations*".

One of the main challenges for environmental management of the seabed is the considerable lack of data to determine ecological thresholds to support decision-making. Hence, the development of key metrics that may serve as threshold ecological indicators are needed [43]. Policy options regarding the exploration and exploitation of marine mineral and biological resources are recommended to balance choices between economic, social and environmental aspects associated with future developments [41].

5. Conclusions

To answer the question, "how can marine ES support the Blue Growth agenda?" the analysis summarized in Fig. 1 shows that the demand for ES is social and economically driven, and in the case of the Blue Growth agenda, there is the demand for: fish protein for human consumption; active compounds for nutraceutics, cosmetics and pharmaceutics; reduction of the dependency on non-renewable abiotic energy sources; cultural experiences; and demand for rare earth elements, commonly used in industrial metals and non-renewable energy.

Marine ES supply biomass from plants and animals for marine aquaculture, and biomass from all biota for direct use or processing for biotechnology. Both ES and the abiotic outputs from seas and ocean supply energy from algae biomass or from renewable abiotic sources (blue energy). They also provide interactions with environmental or with physical settings for maritime, coastal and cruise tourism. Abiotic outputs supply abiotic substances, materials and energy as marine mineral resources. However, the capacity for marine ecosystems to supply the required services in a sustainable way requires actions that are regulated by legislation. In this way human well-being can be linked to GES. To conclude, blue growth requires navigating trade-offs between economic, social and environmental aspects underpinned by marine ES.

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[1] For example in the European Union (Galicia-Northwest of Spain) http://www.farodevigo.es/opinion/2016/11/13/creciente-invasion-china-desafia-pesca/1569184.html

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